PROJECT AND TEAM INFORMATION

Project Title

Campus Skill Swap Network

Student / Team Information

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Project Summary

Campus Skill Swap Network is a C++ based platform where students can find and exchange skills with each other. Students can add the skills and they can teach their skills and can learn also. The system matches students based on skills , mode , location , and rating . DSA concepts like searching , sorting , and some other data structures will be used to make the matching process efficient.

Keywords

Campus Skill Exchange, Peer-to-Peer learning, Student networking, Skill matching, C++, Data structures, searching, sorting, student profiles, online learning, offline learning, compatibility matching, rating system, skill sharing.

PROPOSAL DESCRIPTION

Motivation

Students have different skills, but finding the right person to learn from on campus can be difficult. Many students depend on friends, classmates, or random groups to find someone who can teach a particular skill. There is no simple system to connect students based on what they can teach and what they want to learn.

Our project aims to make this process easier by creating a Campus Skill Swap Network, where students can connect with each other and exchange skills. This can help students learn from their peers while also sharing the skills they already know.

State of the Art / Current Solution

Currently, students usually find people to learn skills from through friends, classmates, college groups, or social media. Online learning platforms also provide many courses and tutorials, but they do not focus on direct skill exchange between students on the same campus.

These methods can help students learn, but finding a suitable person based on skill, learning mode, location, and availability can still take time. Our project aims to bring these things together in one simple system and provide suitable matches between students.

Project Goals and Milestones

Project Goals

- Create student profiles with their skills and preferences.
- Match students based on required and offered skills.
- Consider online/offline mode and location.
- Rank suitable matches using DSA concepts.

- Allow students to send and manage skill exchange requests.
- Provide a simple way to connect learners with suitable mentors or peers.
- Encourage students to share their knowledge and learn new skills from others.

Initial Milestones

1. Requirement and system design
2. C++ class and data structure implementation
3. Student and skill management
4. Matching and ranking system
5. Request and rating system
6. Test the system and evaluate the result.

PROPOSED SOLUTION

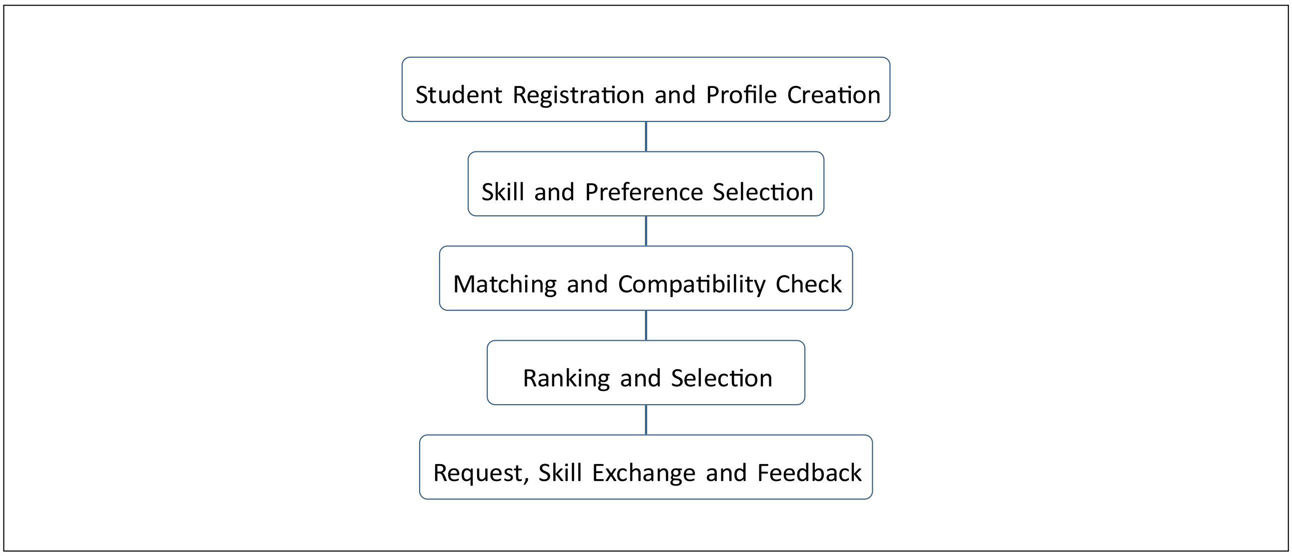
Project Approach

The project will be developed in C++ using object-oriented programming and DSA concepts. First, students will create their profiles by adding the skills they can teach the skills they want to learn, preferred mode, and location.
The system will search for suitable students based on these details and find the best skill matches. Searching and sorting techniques will be used to improve the matching and ranking process. Students will then be able to view suitable matches, send requests and accept requests for skill exchange. The system will also include ratings to help students choose reliable matches.

Key Features

- Student registration and profile creation
- Add skills to teach and skills to learn
- Online, offline, or both mode selection
- Skill-based student/mentor matching
- Compatibility score and match ranking
- Send, accept, and manage skill exchange requests
- Student ratings and feedback
- Simple and user-friendly interface
- Use of C++ OOP and DSA concepts

System Flow



SYSTEM PROCESS AND PROJECT OUTCOME

System Flow Description

1. Registration and Profile Creation – Students register on the platform and create a profile by adding their basic information.
2. Skill and Preference Selection – Students mention the skills they can teach, the skills they want to learn, and select their preferred mode such as online, offline, or both.
3. Matching and Compatibility Check – The system searches for students who have the required skills and checks factors like skill match, preferred mode, and location.
4. Ranking and Selection – The suitable matches are ranked according to their compatibility, making it easier for the student to choose the most suitable person.
5. Request, Skill Exchange and Feedback – The student sends a request to the selected match. After acceptance, both students can exchange skills and provide feedback or ratings after the interaction.

Project Outcome / Deliverables

The project will provide a simple platform for students to find suitable peers for learning and exchanging skills. It will make skill discovery and matching easier while demonstrating the use of C++, OOP, and DSA concepts in a practical application.

Expected Outcomes

- A working Campus Skill Swap Network in C++.
- Easy student registration and profile management.
- Skill-based matching between learners and mentors.
- Ranking of suitable matches using DSA concepts.
- Skill exchange request and feedback system.
- Practical implementation of C++, OOP, and DSA concepts.

Project Deliverables

- Complete C++ source code for the Campus Skill Swap Network.
- Student registration and profile management module.
- Skill matching and compatibility ranking module.
- Request, skill exchange, and feedback module.
- Final project report and documentation.

Assumptions

The project assumes that the users are students from the same campus and provide correct information about their skills, learning preferences, and location. A student can act as both a learner and a mentor. The system also assumes that students have access to the platform whenever they want to find or exchange skills.

Technology Stack

Programming Language	C / C++
Data Structures	Arrays, Linked Lists, Queues
Object-Oriented Concepts	Classes, Objects, Encapsulation, STL
Development Environment	Visual Studio Code
Version Control	Git / GitHub

REFERENCES

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2. W3Schools – C++ Programming Basics, *C++ Reference – Standard Template Library (STL)*, 3. College DSA notes and classroom material